

Program : Diploma in Printing Technology	
Course Code : 3105	Course Title: Color Analysis Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide tangible experience for the students with the basic notions of color reproduction and color analysis practices.
- To develop concrete knowledge on the various aspects of color image reproduction.
- To acquire precise skill in -basic measurement of color & density, color management practices, print quality measurement and analysis,
- To familiarization with various color reproduction quality standards, characterization targets, test charts etc.

Course Prerequisites:

Topic	Course code	Course name	Semester
Visible spectrum, Lightness, Brightness		Applied Physics I	I

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Levels
CO1	Preparing Originals and Measuring Color & Density of print.	10	Applying
CO2	Apply Color Management process.	13	Applying
CO3	Apply Print Quality Measurement.	13	Applying
CO4	To work with Color Reproduction Quality Standards.	5	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Preparing Originals and Measuring of Color & Density of print.		
M1.01	Familiarizing with half tone, continuous tone and line copy originals.	3	Applying
M1.02	Color measurement of CMYK (25%, 50%, 75% & 100%) patches.	2	Applying
M1.03	Density measurement CMYK (25%, 50%, 75% & 100%) patches.	2	Applying
M1.04	Delta E measurements and evaluation of different samples.	3	Applying
CO2	Perform Color Management process		
M2.01	Familiarizing with color management software.	1	Applying
M2.02	Familiarizing with color models-RGB, CMY, and CIE LAB.	2	Applying
M2.03	Calibration of Monitor, Scanner, Printer.	3	Applying
M2.04	Familiarizing with Characterization Targets.	3	Applying
M2.05	Characterization of Monitor, Scanner, Printer.	2	Applying
M2.06	Gamut Mapping and Evaluation of Monitor, Scanner, Printer and Print.	2	Applying
	Lab Exam I	2	
CO3	Perform Print Quality Measurement.		
M3.01	Measuring Hue error, Contrast, Ink trapping Grayness.	4	Applying
M3.02	Dot area measurement.	3	Applying
M3.03	Assessment of effect of screen angles and dot shape on color.	3	Applying
M3.04	Measuring and Evaluating dot gain.	3	Applying

CO4	To work with Color Reproduction Quality Standards		
M4.01	Familiarizing with ISO, G7, GATF, FOGRA standards.	3	Applying
M4.02	Familiarizing of Color control charts- Fogra Charts & GATF Charts.	2	Applying
	Lab Exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	Mark D. Fairchild, “ Color Appearance Models ”, John Wiley & Sons Ltd, England, 2005.
R1	Constance Sidles, Rick Sutherland and Barb Karg, “ Graphic Designer's Print & Color Handbook ”, Rockport Publishers Inc, USA, 2005.
R2	Gaurav Sharma, “ Digital Color Imaging Handbook ”, CRC Press, USA, 2003.
R3	Lindsay W. MacDonald and M. RonnierLuo, “ Color Image Science ”, John Wiley & Sons Ltd, England, 2002.
R4	Asim Kumar Roy Choudhury, “ Modern Concept of Color and Appearance ”, Oxford & IBH Publishing Co Pvt Ltd, New Delhi, 2000.
R5	Gary G. Field, “ Color and Its Reproduction ”, GATF Press Pittsburgh, USA, 1999.

Online Resources:

Sl. No	Website Link
1	https://en.wikipedia.org/wiki/Color
2	https://last.hit.bme.hu/download/firtha/video/Colorimetry/Hunt_Reproduction
3	http://printwiki.org/Color
4	https://www.xrite.com/page/print-packaging-color-measurement
5	https://whattheythink.com/articles/35761-color-measurement-operation